

In-Class Exercise/Demo



In-Class Demo

- Let's practice the [Nodes and Data Structure with Big O](#).
- First, create a new Github repository [m5-w4-d2-demo](#) and then clone this in your WESTCLIFF folder.
- Next, head over to GAP and download [m5-w4-d2-demo.zip](#) folder, unzip and drop the contents into the clone folder.
- Launch VS Code and navigate to the clone folder.
- Open [demo1.html](#) and preview on browser using Go Live.



In-Class Demo #1

- You will see something like this.
- Let's review the `HTML`
- We have `h1`, `h2`, `p`, `ul` and `li` tags
- Uncomment to activate all three const variable scripts at the bottom.
- Next we will use our browser console to manipulate and exercise with DOM
- Open you console and type in `p.parentNode;`

Shark World

The world's leading source on	shark	related information.
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Types of Sharks

- Hammerhead
- Tiger
- Great White

Make sure to screenshots of your console and changes on HTML page.



In-Class Demo

- You will see something like this.
- We are asking for DOM to give us parent node for p. Which is `<body>`
- Next let's type in `p.parentNode.parentNode;`
- It should give you `HTML`
- Next let's type in `ul.childNodes;`
- It will give you all the `li` with texts

The screenshot shows a web browser window with a page titled "Shark World". The page content includes a header "Shark World", a sub-header "Types of Sharks", and a search bar with the text "The world's leading source on shark related information." Below the search bar, there is a section titled "Types of Sharks".

The browser's developer console is open, showing the following commands and results:

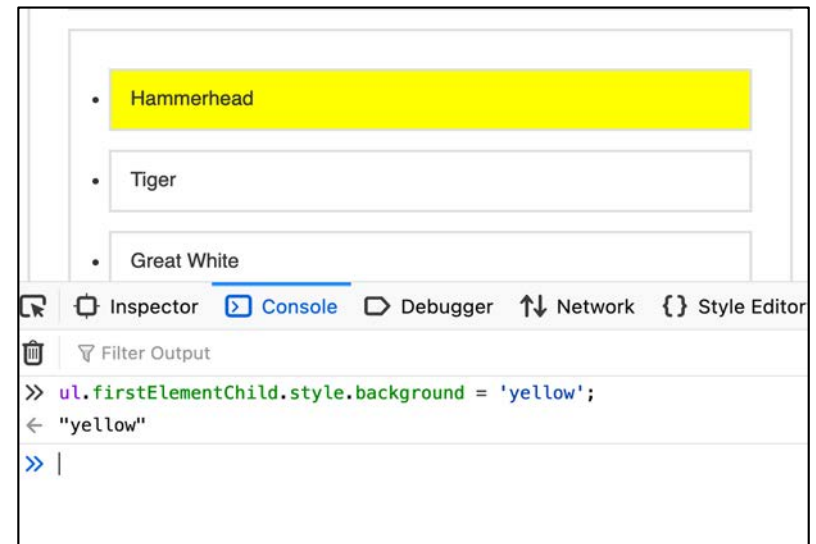
```
>> p.parentNode
< > <body data-new-gr-c-s-check-loaded="8.874.0" data-gr-ext-installed="" cz-shortcut-listen="true">
>>
< > <body data-new-gr-c-s-check-loaded="8.874.0" data-gr-ext-installed="" cz-shortcut-listen="true">
>> p.parentNode.parentNode;
< > <html>
>> ul.childNodes;
< > NodeList(7) [ #text, li, #text, li, #text, li, #text ]
>>
```

Make sure to screenshot of your console and changes on HTML page.



In-Class Demo

- Next, we will change colors.
- Let's type
`ul.firstElementChild.style.background = 'yellow';`
- We asked the ul's first child to have a yellow background



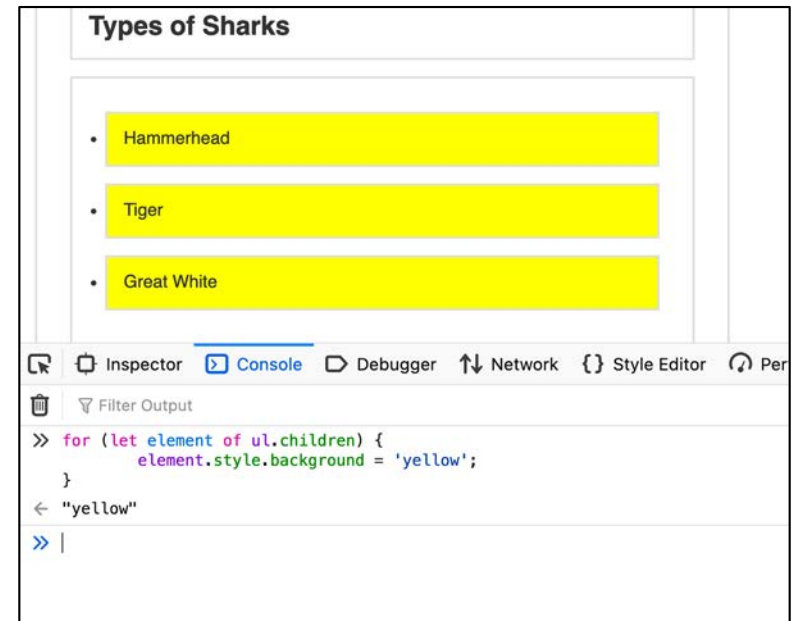
Make sure to screenshot of your console and changes on HTML page.



In-Class Demo

- Next, we will write a for loop to set all the ul's children to yellow
- Enter:

```
for (let element of ul.children) {  
    element.style.background='yellow';  
}
```



Make sure to screenshot of your console and changes on HTML page.



In-Class Demo

- Next, Let's print all the p.childNodes using for loop.
- Enter:

```
for (let element of p.childNodes) {  
    console.log(element);  
}
```

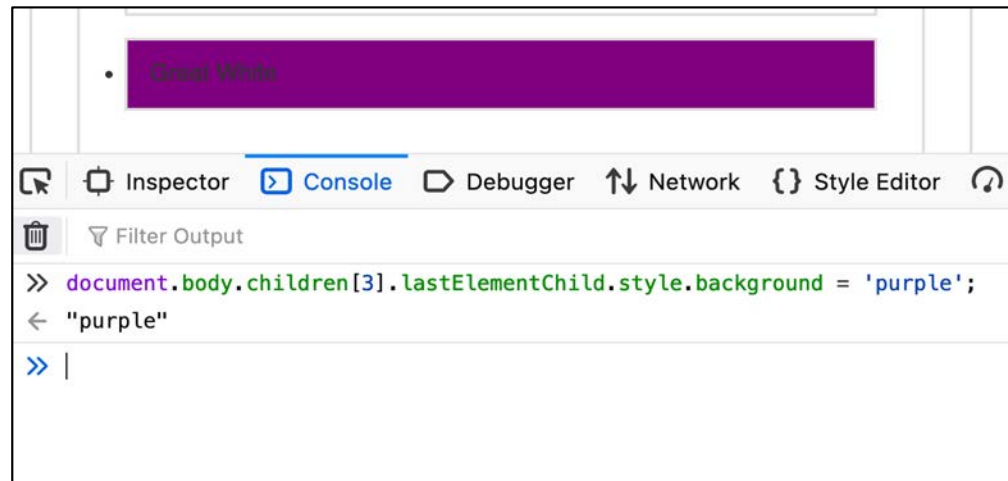
```
>> for (let element of p.childNodes) {  
      console.log(element);  
    }  
▶ #text "The world's leading source on " 🔍  
▶ <strong> 🔍  
▶ #text " related information." 🔍
```

Make sure to screenshot of your console and changes on HTML page.



In-Class Demo

- Next, let's use document array notation to get 1 specific value to change color
- Enter `document.body.children[3].lastElementChild.style.background = 'purple';`



Make sure to screenshot of your console and changes on HTML page.



In-Class Demo #2

- Next, open [demo2.html](#)
- In this example we will use $O(\log n)$ and Quicksort
- Let's open the page in browser and then open your console. We currently do not see anything as we have not add anything to page or to console.
- Go through the code and try to understand what is happening.
- Now add 2 arrays [smArr](#) and [bigArr](#) and add some values to it.

```
const smArr = [37, 29, 30];  
const bigArr = [43, 59, 29, 18];
```



In-Class Demo #2

- Next, uncomment to activate `console.log(sort(smArr));`
- And `console.log(sort(bigArr));`
- Review your console
- The sort method sort the numbers in ascending order

```
► Array(3) [ 29, 30, 37 ]
```

```
► Array(4) [ 18, 29, 43, 59 ]
```

Make sure to screenshot of your console and changes on HTML page.



In-Class Demo #3

- Next, open `demo3.html`
- In this example we will use $O(n!)$
- Let's open the page in browser and open your console. We currently do not see anything as we have not add anything to page or to console.
- Go through the code and try to understand what is happening.
- Now uncomment to activate the following to call the function and then `console.log` it:

```
console.log(factorial(1));
```



In-Class Demo #3

- You will notice that the **factorial** was done really quickly. **2 milliseconds** to be precise.
- Now, let's get a little complicated. Uncomment the following:

```
console.log(factorial(10));
```
- It will take about **85 milliseconds** for this to print.
- As we ask for **higher factorial**, our code is taking longer time to do the Big O
- Now let's add a more complicated factorial. Uncomment the following:

```
console.log(factorial(12));
```
- You will notice this will take an even longer time than others.



In-Class Demo #3

- You will see this output in your console.

1
3628800
479001600
>>

Make sure to screenshot of your console and changes on HTML page.



In-Class Demo #4

- Next, open [demo4.html](#)
- We are going to practice **linear $O(n)$** which is a basic for loop function.
- We will pass the **n number** to the function and it will give us a result in a linear format.
- Enter:

```
exampleLinear(10);
```

Open the console:



0
1
2
3
4
5
6
7
8
9

Make sure to screenshot of your console and changes on HTML page.



In-Class Demo #5

- Next, open [demo5.html](#)
- We are going to practice [deleting](#) an element.
- Open the page in [live preview](#)
- Let's add a [person](#) to the document.

```
document.getElementById("demo").innerHTML =  
person.firstname + " is " + person.age + " years old.";
```

We will delete object properties with the delete keyword.

West is 80 years old.

Make sure to screenshot of your console and changes on HTML page.



In-Class Demo #5

- Now let's **delete** one of the properties.
- Add `delete person.age;` below `document.getElementById.....` code block
- Refresh the page. You will see nothing changed.
- Think for a **second why nothing changed**. We added the values to document and then asked DOM to **remove age**. Now, if you were to check `person.age` in the console. It will tell you it doesn't exist. (**undefined**)
- Let's move `delete person.age;` above the `document.getElementById.....` code block

```
delete person.age;  
document.getElementById("demo").innerHTML =  
person.firstname + " is " + person.age + " years old.";
```




In-Class Demo #5

- As the page is refreshed, notice that age shows `undefined`.

We will delete object properties with the delete keyword.

West is undefined years old.

Make sure to screenshot of your console and changes on HTML page.



In-Class Exercise Demo

Due:

Today 10.30PM PT

Submission:

Drop all your screenshots on to the clone folder and push to your repository.

Post your Github URL link on GAP Week 4 Day 2 Exercise